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RESEARCH NOTE
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Telling a change in transmission from a change in reporting

A POSSIBLE RESEARCH DIRECTION

When reported cases rise, has transmission changed, or only reporting? Which extra measurement would tell us, how big does it need to be, and by when does it have to be taken?

APPROACH

1. **Small cases where the right answer can be computed.** I take stochastic outbreak models small enough for exact likelihoods, with a reporting rate that is unknown and may change, and compute how well any test could do with and without an extra measurement. This gives a benchmark.
2. **Learning what to measure next.** A policy that decides, under a budget, when to survey and how many people to test. It is trained on simulated outbreaks, scored against the benchmark, then moved to a realistic simulator where exact answers are out of reach. MetaWards with a reporting layer might be that simulator.

WHERE THIS COULD GO

FIRST. A map for small closed populations of when a survey has to be taken to be worth taking, with one real episode as a case study.

THEN. A learned policy for what to measure next, with an open benchmark so that other methods can be compared on it.

LONGER TERM. The same question arises wherever a signal and its measurement can both change, for example when monitoring AI systems whose detectors drift.

CLOSEST WORK I HAVE FOUND SO FAR

Bergström, Favero and Britton (2026): in an SIR model, under-reporting, prior immunity and transmission cannot be estimated together from reported cases, and one sample survey fixes this. There under-reporting is a fixed fraction; here it changes, and survey timing is the question.

WHAT IT DRAWS ON

Infectious disease modelling and surveillance data; statistical decision-making (what can be decided from data, and at what cost); machine learning for sequential decisions.

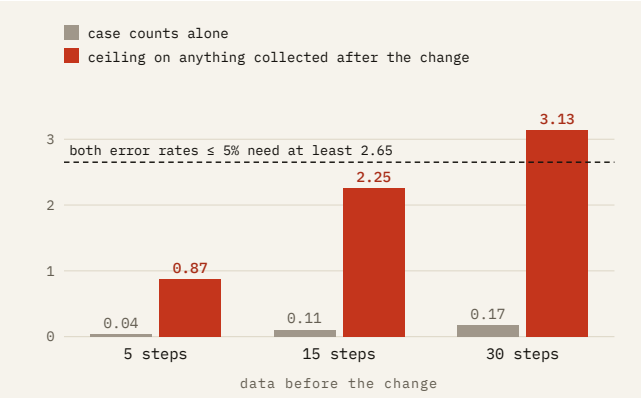
BACKGROUND

First-year PhD student in Engineering Mathematics at Bristol. MSc and PhD work on stochastic processes (first-passage problems); a first-author paper on evaluating LLM agents accepted at UKAIRS 2026.

WHAT I HAVE SO FAR

One scenario: a stochastic SIS model of 60 people. Code written with AI coding tools and checked against a second implementation.

- **Unknown reporting is costly.** With the reporting rate known, the best possible test on a single outbreak is wrong about 20% of the time. With an unknown reporting rate that may change once, about 43%.
- **Timing matters more than size.** With 15 time steps of data before the change (three infectious periods), no prevalence survey taken after the change, however large, can bring both error rates under 5%. With 30 steps this limit no longer binds.
- **More tests soon stop helping.** Sampling 20 of the 60 after the change is worth nearly as much as testing everyone.



Evidence in nats of Kullback–Leibler divergence. The ceiling is a numerical upper limit on everything that can be learned after the change. Clearing the dashed line does not mean it can be done.

WHERE IT MAY NOT MATTER

A rough estimate says that with about 10,000 people or more, and only one reporting change allowed, the shape of the case curve may be enough on its own. So the hard cases should be small closed populations (care homes, wards, schools) or periods when testing policy kept changing. Not covered yet: immunity and serology, reporting delays, over-dispersion.